KAIZEN MATHS

Lesson starter

Upper and Lower Bounds

Error Intervals

Name _____

S1

Find the requested bound for each rounded quantity.

A measurement is 8.9 cm to the nearest 0.1. Find the upper bound.

S2

Write the full error interval for each rounded quantity.

A value x is 50 kg to the nearest 10. Write the error interval for x.

S3

Use the error interval to identify the rounded value and accuracy.

A value x has error interval $55.95 \text{ m} \leq x < 56.05 \text{ m}$. State the rounded value and the accuracy.

S4

Find the requested bound for each rounded quantity.

A measurement is 41.2 kg to the nearest 0.1. Find the upper bound.

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KAIZEN MATHS

Exit ticket

Upper and Lower Bounds

Error Intervals

Name

EA1

Write the full error interval for each rounded quantity.

A value x is 42 kg to the nearest whole number. Write the error interval for x.

EA2

Use the error interval to identify the rounded value and accuracy.

A value x has error interval 78.5 kg ≤ x < 79.5 kg . State the rounded value and the accuracy.

Show your working

KAIZEN MATHS

Exit ticket

Upper and Lower Bounds

Error Intervals

Name

EB1

Find the requested bound for each rounded quantity.

A measurement is 52.7 g to the nearest 0.1. Find the lower bound.

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Exit ticket

Upper and Lower Bounds

Error Intervals

Name

EA1

Write the full error interval for each rounded quantity.

A value x is 42 kg to the nearest whole number. Write the error interval for x.

EA2

Use the error interval to identify the rounded value and accuracy.

A value x has error interval 78.5 kg ≤ x < 79.5 kg . State the rounded value and the accuracy.

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KAIZEN MATHS

Exit ticket

Upper and Lower Bounds

Error Intervals

Name

EB1

Find the requested bound for each rounded quantity.

A measurement is 52.7 g to the nearest 0.1. Find the lower bound.

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Teacher answers & working

Upper and Lower Bounds · Error Intervals

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S1

Find the requested bound for each rounded quantity.

A measurement is 8.9 cm to the nearest 0.1. Find the upper bound.

Answer:

8.95 cm

Working:

The rounding unit is 0.1 cm.

Half of the rounding unit is 0.05 cm.

Lower bound = 8.9 - 0.05 = 8.85.

Upper bound = 8.9 + 0.05 = 8.95.

Upper bound = 8.95 cm.

S2

Write the full error interval for each rounded quantity.

A value x is 50 kg to the nearest 10. Write the error interval for x.

Answer:

$45 \text{ kg} \leq x < 55 \text{ kg}$

Working:

The rounded value is 50 kg.

The rounding unit is 10 kg, so half the rounding unit is 5 kg.

Lower bound = 50 - 5 = 45.

Upper bound = 50 + 5 = 55.

The upper bound is not included because values reaching it would round to the next value.

Therefore 45 kg ≤ x < 55 kg.

S3

Use the error interval to identify the rounded value and accuracy.

A value x has error interval 55.95 m ≤ x < 56.05 m . State the rounded value and the accuracy.

Answer:

56 m, to the nearest 0.1

Working:

Find the midpoint of the interval.

Midpoint = $\frac{55.95+56.05}{2}$ = 56.

Interval width = 56.05 - 55.95 = 0.1.

The interval width is the rounding unit.

So x was rounded to 56 m, to the nearest 0.1.

Teacher answers & working

Upper and Lower Bounds · Error Intervals

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S4

Find the requested bound for each rounded quantity.

A measurement is 41.2 kg to the nearest 0.1. Find the upper bound.

Answer:

41.25 kg

Working:

The rounding unit is 0.1 kg.

Half of the rounding unit is 0.05 kg.

Lower bound = 41.2 - 0.05 = 41.15.

Upper bound = 41.2 + 0.05 = 41.25.

Upper bound = 41.25 kg.

EA1

Write the full error interval for each rounded quantity.

A value x is 42 kg to the nearest whole number. Write the error interval for x.

Answer:

$41.5 \text{ kg} \leq x < 42.5 \text{ kg}$

Working:

The rounded value is 42 kg.

The rounding unit is 1 kg, so half the rounding unit is 0.5 kg.

Lower bound = 42 - 0.5 = 41.5.

Upper bound = 42 + 0.5 = 42.5.

The upper bound is not included because values reaching it would round to the next value.

Therefore 41.5 kg ≤ x < 42.5 kg.

EA2

Use the error interval to identify the rounded value and accuracy.

A value x has error interval 78.5 kg ≤ x < 79.5 kg . State the rounded value and the accuracy.

Answer:

79 kg, to the nearest whole number

Working:

Find the midpoint of the interval.

Midpoint = $\frac{78.5+79.5}{2}$ = 79.

Interval width = 79.5 - 78.5 = 1.

The interval width is the rounding unit.

So x was rounded to 79 kg, to the nearest whole number.

Teacher answers & working

Upper and Lower Bounds · Error Intervals

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EB1

Find the requested bound for each rounded quantity.

A measurement is 52.7 g to the nearest 0.1. Find the lower bound.

Answer:

52.65 g

Working:

The rounding unit is 0.1 g.

Half of the rounding unit is 0.05 g.

Lower bound = 52.7 - 0.05 = 52.65.

Upper bound = 52.7 + 0.05 = 52.75.

Lower bound = 52.65 g.